UNIT 2

Activity 3: Sentiment Analysis Pipeline with Social Media Data

Objective

This activity strengthens student capability in building an end-to-end sentiment analysis pipeline for large-scale social media datasets.The activity introduces real-time data acquisition, NLP-based text processing, distributed execution, and scalability handling. Through this activity, the student is able to integrate APIs, preprocess text, train sentiment models, evaluate performance, and interpret insights for large datasets.

Tasks to Perform

Task 1: Social Media Data Acquisition

This task introduces the process of gathering large-volume text data from platforms such as Twitter or Reddit.
The student is able to connect APIs, authenticate access, stream or batch-collect posts, and prepare raw data for further analysis.

Requirements

- Use Tweepy (Twitter) or PRAW (Reddit) for authenticated data collection
- Gather a sufficiently large dataset for analysis
- Organize outputs into structured formats such as CSV or JSON

Task 2: Text Preprocessing and NLP Preparation

This task focuses on preparing raw social media text for sentiment modeling. The student is able to clean, tokenize, normalize, and transform text using NLTK, TextBlob, or other NLP resources.

Requirements

- Apply tokenization, stopword removal, normalization
- Handle emojis, hashtags, URLs, and slang
- Prepare labeled or pseudo-labeled text for modeling

Task 3: Sentiment Analysis Modeling

This task builds skill in implementing sentiment classification using rule-based or machine learning–based techniques. The student is able to create sentiment models with TextBlob, VADER, ML classifiers, or distributed Spark ML pipelines.

Requirements

- Implement at least one sentiment analysis method
- Experiment with multiple preprocessing variations
- Compare accuracy or F1-scores across model versions

Task 4: Scalability and Distributed Processing

This task strengthens the student's understanding of running NLP tasks on large datasets efficiently. The student is able to use Spark or parallel Python techniques to scale analysis and monitor execution behavior.

Requirements

- Process data with distributed frameworks such as Spark RDDs or DataFrames
- Apply caching or partitioning where needed
- Record throughput or performance observations at scale

Task 5: Insights, Interpretation, and Ethical Considerations

This task builds the student's ability to derive meaningful insights from sentiment outputs. The student is able to interpret trends, generate visualizations, and acknowledge ethical responsibilities in handling social data.

Requirements

- Produce insights such as temporal patterns or topic-based sentiment trends
- Create visualizations such as word clouds, bar charts, or heatmaps
- Address privacy, anonymity, fairness, and bias awareness

What Students Must Submit

Students submit a complete set of artifacts demonstrating the full sentiment analysis workflow. Through these submissions, the student is able to demonstrate technical understanding and clear documentation skills.

Submission Components

- Data acquisition script using Tweepy/PRAW
- Preprocessing and modeling code files
- Accuracy comparison or evaluation table
- Scalability notes and performance observations
- Visualizations and interpreted insights
- Ethical reflection on privacy and bias
- Final project folder or GitHub repository link

Learning Outcome

Upon completing this activity, the student is able to:

- Acquire and manage large-scale social media datasets
- Preprocess text using NLP techniques suitable for sentiment analysis
- Build sentiment models and compare performance across approaches
- Execute distributed analytics pipelines and manage scalability challenges
- Derive meaningful insights supported by visual representations
- Recognize ethical responsibilities in social data processing

Rubric – Sentiment Analysis Pipeline

Task	Excellent (A student will achieve this level when…)	Good (A student will achieve this level when…)	Satisfactory (A student will achieve this level when…)	Poor (A student will achieve this level when…)	No Attempt (0)
Data Acquisition	Identifies API constraints, explains rate-limit handling, analyzes scaling behavior, and evaluates parallel acquisition to produce large, clean datasets.	Identifies API steps, explains basic limit handling, and analyzes completeness for moderate datasets.	Identifies simple API calls, explains minimal handling, and analyzes small datasets with limited stability.	Identifies few acquisition steps, explains errors weakly, and analyzes incomplete or corrupted data.	No acquisition performed.
Analysis Techniques	Applies advanced NLP pipelines, analyzes text deeply, evaluates sentiment accuracy, and justifies model choices using distributed processing.	Applies solid NLP methods, analyzes sentiment reliably, and explains accuracy moderately.	Applies basic NLP rules, explains simple outputs, and analyzes sentiment with minimal depth.	Applies incorrect NLP techniques, explains outputs poorly, and analyzes results inaccurately.	No analysis performed.
Scalability & Optimization	Analyzes distributed workloads, evaluates throughput, applies caching/partition tuning, and justifies improvements using performance metrics.	Analyzes performance moderately and applies basic Spark optimizations.	Analyzes small-scale execution with minimal tuning and limited notes.	Analyzes execution poorly, applies no optimization, and demonstrates weak understanding.	No scalability work performed.
Insights & Interpretation	Analyzes trends deeply, evaluates sentiment shifts, creates strong visualizations, and interprets results with evidence.	Analyzes sentiment distributions clearly with adequate visuals.	Provides simple observations with minimal visuals and basic interpretation.	Provides unclear interpretations with weak or missing visuals.	No insights submitted.

Ethical Considerations	Identifies privacy risks, explains fairness metrics, analyzes bias sources, and evaluates mitigation strategies thoroughly.	Identifies ethical issues, explains impacts, and analyzes basic mitigation steps.	Identifies ethics superficially and explains minimal impact.	Provides unclear or incorrect ethical reflection with no analysis.	No ethical considerations included.